Engineering, Design and Society

Degrees

- Master of Science in Humanitarian Engineering and Science (thesis and non-thesis options)
- Graduate Certificate in Humanitarian Engineering and Science

Program Description

The mission of the Department of Engineering, Design, and Society (EDS) is to engage in research, education, and outreach that inspires and empowers engineers and applied scientists to become innovative and impactful leaders in socio-technical design, problem definition, and problem solution. Our graduates are prepared to address the challenges of attaining a thriving, sustainable global society. Humanitarian Engineering and Science (HES) is a set of graduate program offerings within EDS.

HES connects students with a passion for contributing to social and environmental problem solving to Mines faculty who lead the field of applying engineering to pressing social, environmental, and community challenges. Integrating engineering with social sciences and design, HES offers courses that teach students how to work *with* the communities they seek to serve by co-creating solutions that promote justice, responsibility, and sustainability. HES serves students who have diverse career goals spanning NGOs, government agencies and research groups, start-ups, and established companies. Seminar-style courses offered by EDS, along with selected technical courses offered by other academic units across campus, provide students a balance of breadth and depth in applying engineering to social, environmental, and community challenges.

Information on the Humanitarian Engineering and Science degree programs can be found in the Interdisciplinary Programs section of the catalog.

Primary Contact

Humanitarian Engineering and Science Program
https://humanitarian.mines.edu/

Courses

EDNS501. COMMUNITY-CENTERED APPROACHES TO RESILIENCE. 3.0 Semester Hrs.
Community resilience to the inevitable effects of a changing climate is an emmerging topic in local and global community development spheres. This course will provide learners with a survey of how institutions from governments, donors, to public and private organizations are quickly pivoting from a world of ‘sustainable development goals’ to one of resilience, remediation, and adaptation at national, subnational, and local scales, and how sociotechnical perspectives can support their participation in this new world. At the end of this eight-week course, learners will be able to assess and answer the following questions: [1] what is 'community resilience' and how do different communities, stakeholders, and sectors define it, [2] how does new terminology around ‘resilience’ relate to previous humanitarian efforts aimed at sustainable community development, [3] what are significant sociotechnical pillars of resilience, including built infrastructure and the environment, social infrastructure and governance, [4] how can mainstream efforts aimed at resilience lead to inequality and perpetuate systemic power imbalance. Experiential learning through guest speakers and case studies will illuminate the varied and multidimensional and intersectional aspects of resilience.

Course Learning Outcomes

EDNS502. MANAGING AND LEADING IN MULTISTAKEHOLDER ENVIRONMENTS. 3.0 Semester Hrs.
Managing, leading, and dare we say ‘solving’ wicked public interest issues requires multistakeholder engagement across sectors, agencies, and public interest domains. This course is designed to provide learners with tools to manage and lead across these complex multistakeholder environments, and to navigate wicked problems, like climate change and community resilience. This course is designed for those who are or aspire to be public-facing managers and leaders—including those working for state and federal agencies, nonprofits, non-governmental organizations (NGOs), and quasi-governmental organizations. Key learning objectives include [1] defining the public interest, [2] assessing stakeholder environments, [3] exploring ethical decision-making tools and considering competing stakeholder interests, [4] analyzing the functions and roles of the ‘responsible’ public manager, [5] synthesizing modern community-engaged leadership styles and applying them in practice, [6] exploring current challenges and perspectives from leaders in relevant fields and organizations, and [6] learning and applying advanced communication skills.
Course Learning Outcomes

EDNS515. INTRODUCTION TO SCIENCE AND TECHNOLOGY STUDIES. 3.0 Semester Hrs.
This course engages scholarship on the inextricable link between science, engineering and the various social contexts within which scientists and engineers work. We begin by critically reflecting on the question, What are science and engineering for? We then explore key conceptual domains in the social scientific study of science and engineering, including knowledge, agency, and expertise. We will learn from a diverse set of social scientific experts who study and collaborate with scientists and engineers. Students will leave the course with a better understanding of how social scientific inquiry can aid in understanding, and practicing, science and engineering. They will also have a clearer articulation of their individual professional commitments and how those fit with more traditional understandings of science and engineering.
Course Learning Outcomes

- By the end of this course, students will have demonstrated the ability to:
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EDNS540. PROTOTYPING FOR CONCEPT VALIDATION. 3.0 Semester Hrs.
This course develops students’ ability to use prototypes as strategic tools for concept exploration, decision-making, and validation throughout engineering design processes. Rather than treating prototypes as final deliverables, students will conceive, model, and fabricate a series of purpose-driven prototypes to test requirements, functionality, user experience, system integration, and design for manufacturability. Through iterative design cycles, students will explore the trade-offs between materials, fidelity, and media to collect targeted data, solicit stakeholder feedback, assess market fit, and evaluate associations of comparative fabrication and assembly methods’ effects on cost. Emphasis is placed

on selecting appropriate prototype forms at various milestones of engineering problem solving to effectively validate hypotheses and support data-driven design decisions.

Course Learning Outcomes

- Conceive and deploy prototypes to verify or validate specific component or system requirements, design specifications, functions, expected user experiences, effective system integrations, and other granular aspects of a larger solution concept.
- Utilize prototypes to solicit feedback from potential customers or other stakeholders to confirm market fit and desirability.
- Explore the trade-offs between levels of fidelity and various media as they relate to data collection and concept validation.
- Evaluate fabrication methods to assess prototyping costs in relation to manufacturability and projected manufacturing costs.
- Incorporate rapid iteration cycles with minimal viable products (MVPs) to test hypotheses with real-world feedback.
- Ability to critically analyze and apply scholarship to design decision making

EDNS544. INNOV8X. 3.0 Semester Hrs.

Innov8x introduces concepts and tools to accelerate the design, validation and adoption of innovations in support of creative problem solving. Using an entrepreneurial mindset, we learn how to identify and frame problems that beneficiaries and stakeholders face. We attempt to design and test practical solutions to those problems in collaboration with those who experience the problems. We apply beneficiary discovery, pretotyping, business model design (social, economic and environmental), constrained creativity, efficient experimentation, and rapid iteration. While resolving challenges involves technical solutions, an important aspect of this course is directly engaging beneficiaries and stakeholders in social contexts to develop solutions with strong impact potential. Innov8x is grounded in collaborative creativity theory at the intersection of organizational behavior (social psychology), design principles, entrepreneurship and innovation management.

Course Learning Outcomes

- Frame and translate complex ambiguous problems into actionable opportunities for innovation
- Conduct effective, objective and ongoing beneficiary discovery in efficient ways
- Combine tools and methods to quickly test assumptions and secure beneficiary acceptance
- Develop creative approaches to navigate real and perceived constraints
- Leverage mentor and stakeholder support through credible communication based on research • Launch innovative solutions with the advocacy of beneficiaries and stakeholders
- Create value by solving complex problems that straddle technical and social domains
- Launch innovative solutions with the advocacy of beneficiaries and stakeholders

EDNS545. PRODUCT REDESIGN. 3.0 Semester Hrs.

Product redesign reimagines existing products, focusing specifically on a systems approach to human-centered design and the crafting of design solutions tailored to meet the needs of their users. Students will progress through an iterative design process, engaging in the analysis of and thoughtful reflection on design opportunities, ensuring enhanced products align with the needs of a specific user group. Emphasizing collaborative learning, students will work in teams, adopting a multi-disciplinary approach to creative problem-solving and design. Multiple prototyping cycles will guide students as they make data-driven design decisions, culminating in the development and communication of a final redesigned product.

Course Learning Outcomes

- Analyze the needs of a specific group of users in a given context and develop a problem definition that responds to those needs with a clear, concise set of engineering design criteria.
- Create a product design and development plan with a defined timeline that results in an advanced design artifact.
- Propose distinct solution concepts and utilize user feedback, engineering analysis, experimentation, and proven industry practices to make data-driven design decisions.
- Build, test, and analyze solution concepts through a series of design cycles to iterate and refine the advanced artifact.
- Participate equitably on a team with distributed roles and responsibilities, while monitoring individual effectiveness in contributing to the team’s overall progress.
- Ability to critically analyze and apply scholarship to design decision making

EDNS577. ADVANCED ENGINEERING AND SUSTAINABLE COMMUNITY DEVELOPMENT. 3.0 Semester Hrs.

Analyzes the relationship between engineering and sustainable community development (SCD) from historical, political, ethical, cultural, and practical perspectives. Students will study and analyze different dimensions of sustainability, development, and "helping", and the role that engineering might play in each. Will include critical explorations of strengths and limitations of dominant methods in engineering problem solving, design and research for working in SCD. Through case-studies, students will analyze and evaluate projects in SCD and develop criteria for their evaluation. 3 hours lecture and discussion; 3 semester hours.

Course Learning Outcomes

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EDNS578. ENGINEERING AND SOCIAL JUSTICE. 3.0 Semester Hrs.

Equivalent with LAIS578,

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EDNS579. COMMUNITY-BASED RESEARCH METHODS. 3.0 Semester Hrs.

Engineers and applied scientists face challenges that are profoundly sociotechnical in nature, and communities are increasingly calling for greater participation in the decisions that affect them. Understanding the diverse perspectives of communities and being able to establish positive working relationships with their members is therefore crucial to the socially responsible practice of engineering and applied science. This course provides graduate students with the conceptual and methodological tools to conduct community-based research. Graduate students will learn ethnographic field methods and participatory research strategies, and critically assess the strengths and limitations of these through a final original research project related to their ongoing independent research or practicums.

Course Learning Outcomes

- During this course students will learn to:

EDNS580. HUMANITARIAN ENGINEERING AND SCIENCE CAPSTONE PRACTICUM. 3.0 Semester Hrs.
(I, II, S) This course allows students to practice the concepts, theories and methods learned in HES courses with the goal of making relevant their academic training to real world problems. This practicum can be achieved through a number of possibilities approved by HES director, including supervision and/or shadowing in HES-related activities, engaging in a social enterprise where they do problem definition, impact gap analysis and layout a business canvas, and designing and carrying out a project or fieldwork of their own, etc. Prerequisite: EDNS570, EDNS479. 3 hours research; 3 semester hours.
Course Learning Outcomes

- • Identify successful practices for humanitarian projects in real settings (ABET a,h,j) • Determine different ways in which previous humanitarian projects could have been improved to yield more successful technical and social results (ABET a,b,h,j) • Determine effective engineering methods for different humanitarian applications (ABET b,c,h,j) • Work in teams to design, execute and evaluate a project with stakeholders (ABET a,b,c,d,e,j,k) • Gain experience in engaging and communicating with community members and stakeholders (ABET c,d,f,h,i,j,k) • Develop stronger professional communication skills through written assignments, group projects, discussions, presentations, and community engagement (ABET g,f,h,i,j,k)

EDNS590. RISKS IN HUMANITARIAN ENGINEERING AND SCIENCE. 3.0 Semester Hrs.
(I) This course provides students with opportunities to consider the risks related to humanitarian projects—or any projects that effect and involve people. These risks might include things that different scientific and engineering disciplines typically consider, as well as those that may be pertinent to project stakeholder perspectives. Guided by social scientific insights related to risk, students in this class will gain new tools for defining problems in ways that are relevant and appropriate for multiple contexts. Students will read, discuss, and analyze material together and to undertake independent research to deepen their understandings of chosen topics. 3 semester hours.
Course Learning Outcomes

- • Analyze humanitarian science and engineering projects using established evaluation criteria (ABET a,h,j) • Identify the most successful practices for humanitarian science and engineering (ABET a,h,j) • Determine different ways in which previous engineering or scientific projects could have been improved to yield more successful technical and social results (ABET a,b,h,j) • Gain conceptual tools for and experience in engaging and communicating with community members and stakeholders (ABET c,d,f,h,i,j,k) • Develop stronger professional communication skills through written assignments, group projects, discussions, presentations, and community engagement (ABET g,f,h,i,j,k)

EDNS597. SUMMER PROGRAMS. 0-6 Semester Hr.

EDNS598. SPECIAL TOPICS IN ENGINEERING DESIGN & SOCIETY. 0-6 Semester Hr.
(I, II, S) Pilot course or special topics course. Topics chosen from special interests of instructor(s) and student(s). Usually the course is offered only once, but no more than twice for the same course content. Prerequisite: none. Variable credit: 0 to 6 credit hours. Repeatable for credit under different titles.

EDNS598. SPECIAL TOPICS. 0-6 Semester Hr.

EDNS598. SPECIAL TOPICS. 0-6 Semester Hr.

EDNS599. INDEPENDENT STUDY. 0.5-6 Semester Hr.
Individual research or special problem projects supervised by a faculty member, also, when a student and instructor agree on a subject matter, content, and credit hours. Variable credit: 0.5 to 6 credit hours. Repeatable for credit under different topics/experience and maximums vary by department. Contact the Department for credit limits toward the degree. Independent Study form must be completed and submitted to the Registrar.
Course Learning Outcomes

EDNS707. GRADUATE THESES. 1-15 Semester Hr.

Department Leadership

Dean Nieusma, Department Head

Chelsea Salinas, Assistant Department Head; Director of Design Engineering Program

Professors

Juan Lucena, Humanitarian Engineering Director of Undergraduate Programs and Outreach

Jessica Smith

Assistant professors

Elizabeth Reddy, Assistant Director of Humanitarian Engineering and Science Interdisciplinary Graduate Program

Marie Stettler Kleine

Teaching Professors

Yosef Allam, Director of Cornerstone Design Program

Alina Handorean

Teaching Associate Professors

Jack Bringardner

Mirna Mattjik

Mark Orrs

Michael Sheppard

Teaching Assistant Professors

Cynthia Athanasiou

Duncan Davis-Hall

Aubrey Wigner

Professors of Practice

Donna Bodeau

Garrett Erickson

Antonie Vandenberge

Professor Emeritus

Kevin Moore

Staff

Becky Buschke, Program Assistant

Leah Fitzgerald, Stakeholder Relations Manager

Kirsten Kelly, Capstone Administrative Assistant

Julia Roos, Associate Director of Humanitarian Engineering

Kimberly Walker, Department Manager

Electrical Engineering

Degrees Offered

- Professional Online Masters in Electrical Engineering *This degree can be earned by completing the last three certificates listed below plus one additional course.
- Master of Science (Electrical Engineering)
- Doctor of Philosophy (Electrical Engineering)
- Graduate Certificate in Information and Systems Sciences
- Graduate Certificate in Microwave Engineering
- Graduate Certificate in Power and Energy Systems

Program Overview

The Electrical Engineering Department offer the degrees Master of Science and Doctor of Philosophy in Electrical Engineering. These degree programs demand academic rigor and depth yet also address real-world problems.

The Department has three areas of research activity that stem from the core fields of Electrical Engineering: 1) Antennas and Wireless Communications, 2) Information and Systems Science, and 3) Power and Energy Systems. Individual research projects may encompass more than one research area.

Research Areas:

Antennas and Wireless Communications is a research area that builds on the fundamental physics and mathematics of electromagnetic waves and propagation. The research in this area includes design, analysis, optimization, and measurement of antennas, antenna arrays, microwave, millimeter-wave, and terahertz devices. Applications address current academic, industry, and society needs, such as wireless communication systems, radar and remote sensing, and electromagnetic imaging.

Information and Systems Sciences is an interdisciplinary research area that encompasses the fields of control systems, data science, optimization, signal and image processing, compressive sensing, robotics, and mechatronics. Applications can be found in renewable energy and power systems, materials processing, sensor and control networks, bio-engineering, computer vision and pattern recognition, autonomous systems, imaging, intelligent structures, and geosystems.

Power and Energy Systems is focused on both fundamental and applied research in the interrelated fields of conventional electric power systems and electric machinery, renewable energy and distributed generation, energy economics and policy issues, power quality, power electronics and drives. The overall scope of research encompasses a broad spectrum of electrical energy applications including investor-owned utilities, rural electric associations, manufacturing facilities, regulatory agencies, and consulting engineering firms.

Program Details

The Electrical Engineering Department offers master's degrees and a Doctor of Philosophy in Electrical Engineering. The Master of Science program is designed to prepare candidates for careers in industry or government or for further study at the PhD level; both thesis and non-thesis options are available. A Professional Online Master's degree is also available. The PhD degree program is sufficiently flexible to prepare candidates for careers in industry, government, or academia. See the information that follows for full details on these four degrees.

Mines Combined Undergraduate/Graduate Degree Program

Students enrolled in Mines' combined undergraduate/graduate program may double count up to 6 credits of graduate coursework (500-level) to fulfill requirements of both their undergraduate and graduate degree programs. These courses must have been passed with B- or better, not be substitutes for required coursework, and meet all other university, department, and program requirements for graduate credit.

Students are advised to consult with their undergraduate and graduate advisors for appropriate courses to double count upon admission to the combined program.

Prerequisites

Requirements for Admission to EE: The minimum requirements for admission to the MS and PhD degrees in Electrical Engineering are:

- A baccalaureate degree in engineering, computer science, a physical science, or math with a grade-point average of 3.0 or better on a 4.0 scale.
- TOEFL score of 79 or higher (or 550 for the paper-based test or 213 for the computer-based test) for applicants whose native language is not English. In lieu of a TOEFL score, an IELTS score of 6.5 or higher will be accepted.
- For the PhD program, prior research experience is desired but not required.

Admitted Students: The EE department graduate committee may require that an admitted student take undergraduate remedial coursework to overcome technical deficiencies. The committee will decide whether to recommend regular or provisional admission.

Transfer Courses: Graduate-level courses taken at other universities for which a grade equivalent to a B or better was received will be considered for transfer credit with approval of the advisor and/or thesis committee, and EE department head, as appropriate. Transfer credits must not have been used as credit toward a bachelor's degree. For the MS degree, no more than 9 credits may transfer. For the PhD degree, up to 24 credits may be transferred. In lieu of transfer credit for individual courses, students who enter the PhD program with a thesis-based master's degree from another institution may transfer up to 36 hours in recognition of the coursework and research completed for that degree.

Advisor and Thesis Committee: Students must have an advisor from the EE faculty to direct and monitor their academic plan, research, and independent studies. Advisors must be full-time permanent members of the faculty. In this context, full-time permanent members of the faculty